Intro To AI

SHEET 1

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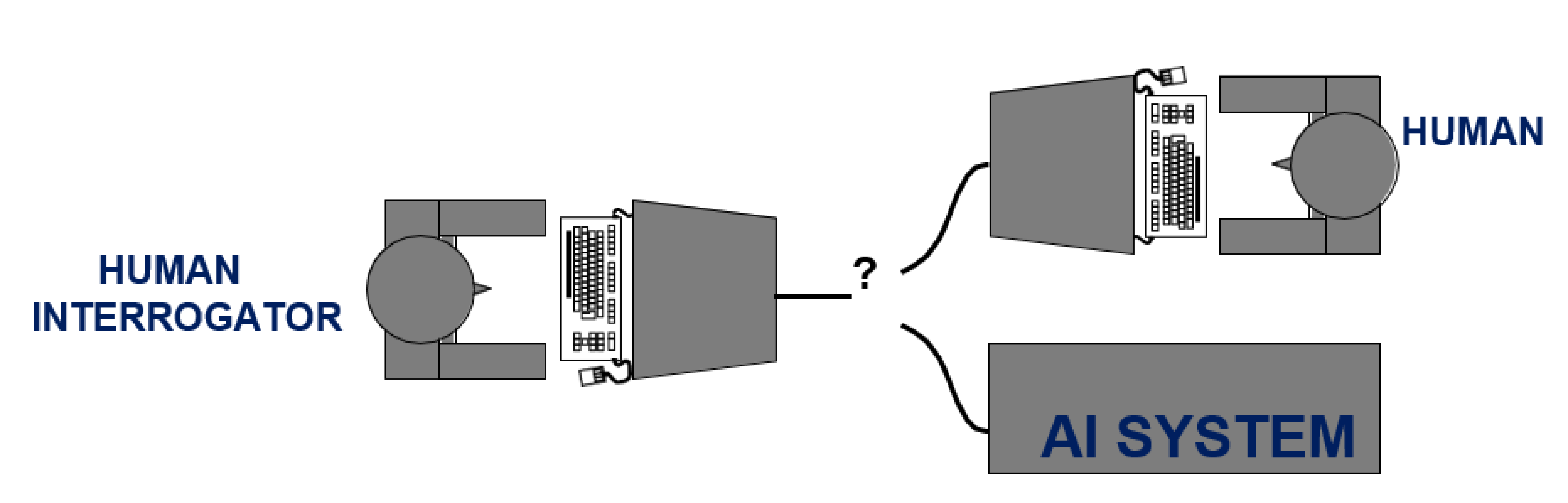
Q.1.a) What is The Turing Test "imitation game". *(Illustrate through drawing)*

Description: Proposed by Alan Turing in his 1950 paper "Computing Machinery and Intelligence", this test replaces the broad question "Can machines think?" with an operational behavioral test: "Can machines behave intelligently?".

Setup: A human interrogator communicates textually with two hidden players—a human and a machine. If the interrogator cannot reliably distinguish between the machine and the human after a series of written questions and answers, the machine passes the test as behaving intelligently.

Q.1.a) What is The Turing Test "imitation game". *(Illustrate through drawing)*



Q.1.b) What is The Total Turing Test.

Description: An extended version of the Turing Test that includes physical aspects of human behavior by evaluating whether a machine can simulate human traits across multiple operational modalities. To pass the Total Turing Test, the system requires:

Natural Language Processing to communicate successfully.

Knowledge Representation to store what it knows or hears.

Automated Reasoning to use stored information to answer questions and draw new conclusions.

Machine Learning to adapt to new circumstances and detect patterns.

Computer Vision to perceive physical objects and surroundings.

Robotics to move and manipulate its physical environment.

Q.1.c) What are Systems that “Act/Behave Rationally”, and Systems that “Act/Behave Humanly”.

Acting Humanly: This approach focuses on making computers do tasks at which people are currently better. Success is measured by how closely a machine copies human actions, imperfections, and cognitive quirks (e.g., passing the Turing Test).

Acting Rationally: This approach drops the requirement to match human behavior. Instead, a rational agent acts to optimally achieve its goals (doing the "right thing") by maximizing its expected performance measure given the available information and resources. This is comparable to building flying machines using aerodynamics rather than exactly mimicking how birds flap their wings.

Q.1.d) What is “Weak AI Hypothesis” versus the “Strong AI Hypothesis”.

Weak AI Hypothesis: Asserts that we can accurately simulate animal or human intelligent behavior in a computer program, acting as an extremely powerful tool for problem-solving without genuine consciousness.

Strong AI Hypothesis: Asserts that we can create algorithms that are genuinely intelligent, possessing actual minds, consciousness, self-awareness, and free-will.

Q.1.e) What are The two most fundamental concerns of AI researchers.

According to the foundational traditions of AI (such as Luger's framework), the two core cross-cutting concerns are:

Knowledge Representation: How to formalize real-world facts, rules, structures, and uncertain domain knowledge into symbolic or numeric forms that a computer can store and manipulate.

Search/Problem Solving: Efficiently navigating a massive graph or space of potential alternative states to discover an optimal sequence of operators leading to a goal.

Q.1.f) What are Intelligent Agents.

Description: An agent is anything that perceives its environment through sensors and acts upon that environment via actuators.

An ***intelligent* agent** selects actions that are appropriate for its goals and circumstances, is flexible to changing environments, and learns from experience to maximize its performance metrics under resource constraints.

Question 2: Give the scientific term for each of the following statements:

Statement	Scientific Term
The branch of computer science that is concerned with the automation of intelligent behavior.	**Artificial Intelligence (AI)**
A problem-solving technique that systematically explores a space of problem states.	**State Space Search**
Systems that are constructed by obtaining knowledge from a human expert and coding it into a form that a computer may apply to similar problems.	**Expert Systems**
Models that parallel the structure of neurons in the human brain and used to build intelligent programs.	**Artificial Neural Networks (ANN)**
Algorithms that evolve new problem solutions from components of previous solutions using specific operators such as crossover and mutation.	**Genetic / Evolutionary Algorithms**

Question 3: Criticize Turing's criteria for computer software being "intelligent"; What is Searle's thought experiment (the Chinese Room Argument)?

Turing's Criteria Criticism

While operational and practical, the Turing Test is criticized because it measures **simulation** rather than **real intelligence**. A program can pass the test using purely syntactic surface-level lookups, brute-force patterns, or deceptive rules without any structural understanding of what it is saying or doing.

Question 3: Criticize Turing's criteria for computer software being "intelligent"; What is Searle’s thought experiment (the Chinese Room Argument)?

Searle’s Chinese Room Argument

The Premise: John Searle asks us to imagine an English-only speaking person locked inside a room with a massive rulebook written in English. This book outlines manual steps for processing incoming Chinese symbols: "If symbol ***X*** comes in, write symbol ***Y*** out".

The Execution: Outside observers pass sheets of paper with Chinese characters into the room through a slot. By following the rulebook instructions, the person writes accurate Chinese responses and passes them back out. To the outside observers, the room passes the Turing Test perfectly as a fluent Chinese speaker.

Question 3: Criticize Turing's criteria for computer software being "intelligent"; What is Searle's thought experiment (the Chinese Room Argument)?

The Conclusion: The person inside is executing symbol manipulation rules blindly and **does not understand a single word of Chinese**. Searle points out that this is exactly what an AI program or computer does: it manipulates abstract symbols syntactically without any semantic comprehension or intentionality. Therefore, passing the Turing Test does not prove strong AI or consciousness.

Thank you